



Plant Biotechnology

Lab. BT 421

Lab (3)

Media Preparation

By:

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Outline

- ☐ **Media composition**
- ☐ **Hormones**
- ☐ **Callus in plants**
- ☐ **Different plant parts induction using MS media**
- ☐ **How to prepare 1 L of MS medium**

Media composition

- Media can be made from the pure chemicals or purchased as commercial premixed culture media.
- Empirical trials may be necessary to test available combinations of ingredients when dealing with a new kind of plant.



Media composition

- Synthesized media must contain:

1. Inorganic salts.

2. Organic compounds.

3. Gelling agents.

1. Inorganic salts

Macronutrients

- ❑ **Nitrogen**: component of proteins, nucleic acid.
- ❑ **Phosphorus**: nucleic acids, ATP.
- ❑ **Potassium**: regulate osmotic potentials.
- ❑ **Calcium**: cell wall synthesis, cell signaling.
- ❑ **Magnesium**: component of chlorophyll.
- ❑ **Sulphur**: component of amino acids and cofactors.

Micronutrients

- ❑ **Manganese + Molybdenum + Copper**: Enzyme cofactor.
- ❑ **Cobalt**: Component of some vitamins.
- ❑ **Boron**: Cell wall synthesis and cell division.
- ❑ **Iron**: Electron transfer.
- ❑ **Zinc**: Chlorophyll biosynthesis.

2. Organic compounds

Carbohydrates

Vitamins

Hormones

(Hormones)/Growth regulators

- ❑ Plant growth regulators are the critical media components in determining the developmental pathway of the plant cells.
- ❑ There are five main classes of plant regulators used in plant tissue culture, namely:
 1. Auxins.
 2. Cytokinins.
 3. Gibberellins.
 4. Absciscic acid.
 5. Ethylene.

(Hormones) and growth regulators



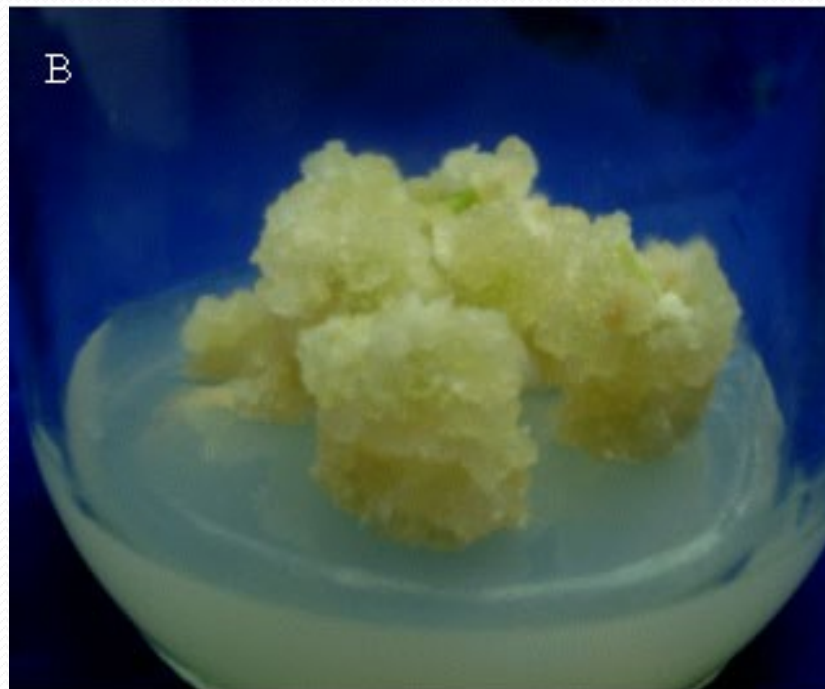
	Germination	Growth to Maturity	Flowering	Fruit Development	Abscission	Seed Dormancy
Gibberellin	✓	✓	✓	✓	✗	✗
Auxin	✗	✓	✓	✓	✗	✗
Cytokinins	✗	✓	✓	✓	✗	✗
Ethylene	✗	✗	✓	✓	✓	✗
Absciscic Acid	✗	✗	✗	✗	✓	✓

3. Gelling agents and PH

- Natural gelling product (Seaweed / Red algae).
- Agar performance may be affected by concentration and by pH.
- 0.5-0.8 % gram (8.0 g /1L).
- Allows adequate movement of nutrients in the medium.
- Pores of the medium makes aeration.
- the solidification of the agar is pH-dependent.

Callus in plants

Plant callus is a growing mass of unorganized plant parenchyma cells. In living plants, callus cells are those cells that cover a plant wound.



Different plant parts induction using MS media

Six different media:

1. MS hormone free medium.

MS salts without hormones.

2. Shooting medium.

MS + 0.5 mg/L **GA₃** + 2 mg/L **BA** + 0.3 mg/L **IBA**.

3. Rooting medium.

MS + 0.5 mg/L **GA₃** + 2 mg/L **IAA**.

Different plant parts induction using MS media

4. Callus A medium.

MS + 0.5 mg/L **GA₃** + 0.5 mg/L **2-4 D**.

5. Callus B medium.

MS + 0.5 mg/L **GA₃** + 2 mg/L **BA** + 0.5 mg/L **IAA**.

6. Water agar medium.

Agar without MS and hormones.

How to prepare 1 L of MS medium:

1. Take a specific amount from each stock solution and then makeup the volume to **1 L** using **D.H₂O**:

50 ml of stock **1.**

5 ml of stock **2.**

5 ml of stock **3.**

5 ml of stock **4.**

How to prepare 1 L of MS medium:

1. Pour **200 ml** of MS stock and **800 ml** D.H₂O in a 1 L-flask putting a magnetic stirring bar.

2. Dissolve **30 g** of sucrose.



****Adding hormones**

3. Adjust the pH to (**5.7-5.9**) using pH meter.

How to prepare 1 L of MS medium:

4. Add **8.0 g** agar and dissolve it by microwave for 5 minutes.
5. Homogenization (magnetic stirrer).
6. Dispense about **15 ml** medium per test tube and cap it with a cotton blog (30 tubes per a rack).

How to prepare 1 L of MS medium:

7. Cover the rack with green proof paper.
8. Autoclave at 121 °C for 15 minutes.
9. Take out from autoclave and allow the medium to cool at room temp.